

Technical Note on Taxable Wage Measures

in Tennessee Administrative Data

EXECUTIVE SUMMARY

This note documents a potential issue with the taxable-wage measure in the employer-level data for 2004–2015 extracted by the Tennessee Department of Labor and Workforce Development (TDLWD) from the Quarterly Census of Employment and Wages (QCEW). During this period, taxable wages equal total wages for each employer-quarter in the matched sample. Because wages are only taxable up to a taxable wage base of \$7,000–\$9,000 per worker for each employer and year, taxable wages should generally be lower than total wages and peak in the first quarter of the year. The recommended solution is to construct employer-year-quarter taxable wages from individual wage records using the applicable annual taxable wage base, and then to aggregate by employer, year, and quarter. This note and the attached replication code are intended as technical resources for TDLWD and other authorized data users. The analysis suggests a reporting problem but does not identify its administrative cause. The proposed solution applies to the specific data extracts provided.

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ON

September 29, 2026

The analysis, interpretations, and any remaining errors in this note are my own. Corrections, alternative interpretations, and information about an existing or improved solution are welcome.

1 Definition of total and taxable wages

For a given employee–employer relationship, total wages are the wages paid during a quarter. Taxable wages are the portion of those wages subject to unemployment–insurance taxation. Within a calendar year, an employee’s wages are taxable only until the taxable wage base is reached for that employer. Wages paid after the threshold is reached are not taxable. The calculation of taxable wages resets to zero in the following calendar year. Taxable wages are therefore generally lower than total wages and tend to be concentrated in the earlier quarters of the year.

The example below uses an annual taxable wage base of \$7,000, the current level in Tennessee, and constant quarterly wages for three workers. Worker 1 earns \$1,000 per quarter, so total and taxable wages are equal throughout the year. Worker 2 earns \$5,000 per quarter: \$5,000 is taxable in Q1 and \$2,000 in Q2, after which the annual wage base has been reached and no additional wages are taxable in that or later quarters. Worker 3 earns \$10,000 per quarter, so \$7,000 is taxable in Q1 and no additional wages are taxable in that or later quarters. As illustrated with this example, total and taxable wages may coincide for low-wage workers, but they are not expected to coincide across all workers, employers, and quarters. Taxable wages also tend to be higher in Q1 than in other quarters.

Worker	Measure	Q1	Q2	Q3	Q4
Worker 1	Total wages	\$1,000	\$1,000	\$1,000	\$1,000
	Taxable wages	\$1,000	\$1,000	\$1,000	\$1,000
Worker 2	Total wages	\$5,000	\$5,000	\$5,000	\$5,000
	Taxable wages	\$5,000	\$2,000	\$0	\$0
Worker 3	Total wages	\$10,000	\$10,000	\$10,000	\$10,000
	Taxable wages	\$7,000	\$0	\$0	\$0

2 Data and sample

The analysis uses two restricted administrative datasets provided by TDLWD.

Dataset	Source	File	Coverage	Unit of observation	Number of observations
Employer	QCEW	Employer_Data_V1.txt	2002–2023	Employer–year–quarter	12,045,233
Employee	Wage Files	BU_UI_Employment.txt	2000–2024	Employee–employer–year–quarter	306,436,346

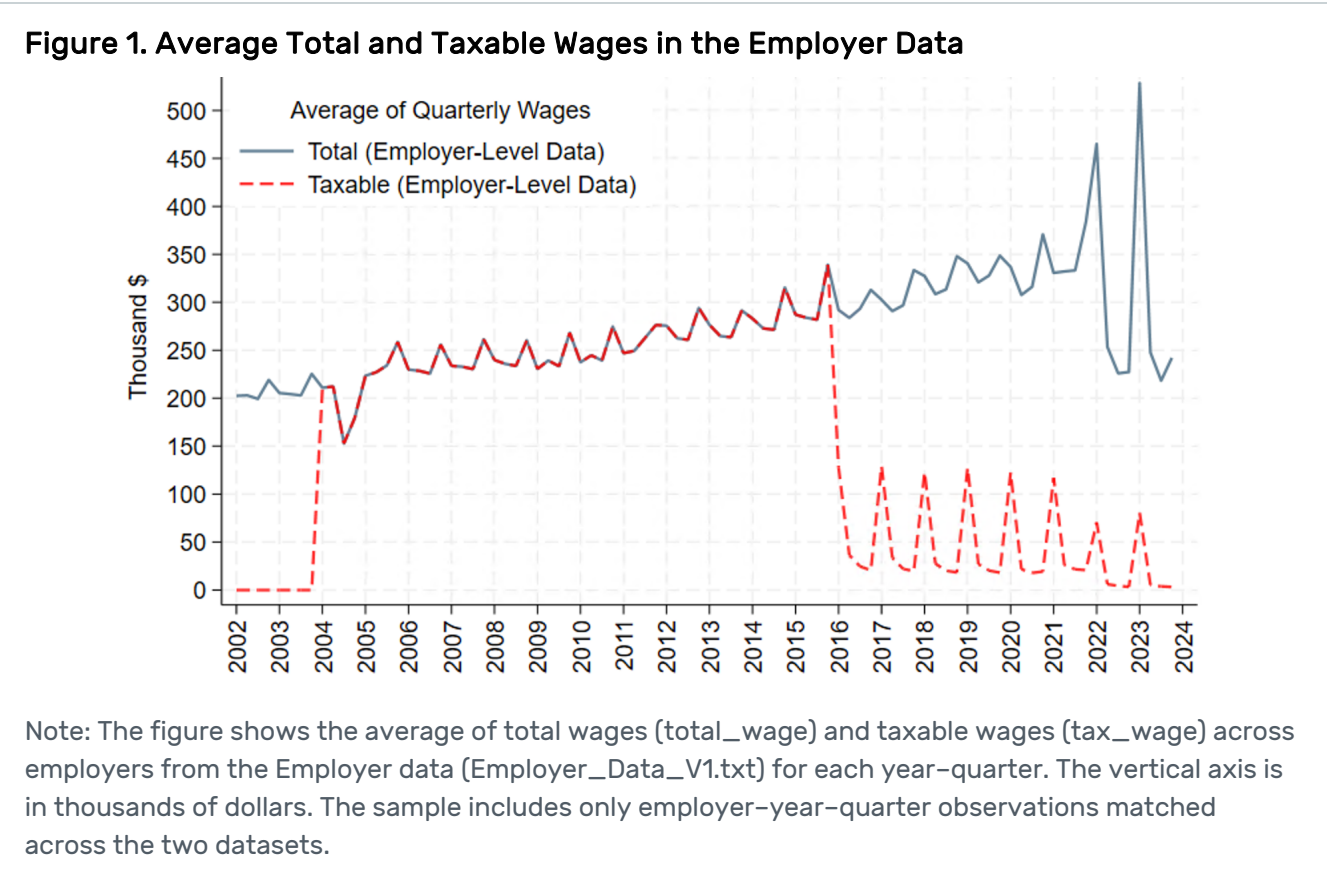
Employer data. Employer–quarter observations with a missing employer identifier or inactive status are excluded. All the employer–quarter observations with duplicates are also dropped. The restricted sample contains 10,492,995 observations, or 87% of the original data. The data contains two key variables: total wages (variable name: total_wage) and taxable wages (variable name: tax_wage). This note evaluates the taxable–wage measure, which equals total wages in every matched observation from 2004 through 2015.

Employee data. The file contains wages at the employee–employer–year–quarter level. First, wages belonging to the same employee–employer–year–quarter are summed. Then, using the applicable annual Tennessee taxable wage base, taxable wages are constructed for each employee–employer–year–quarter. For each employee–employer–year, records are ordered by quarter. Taxable wages in each quarter are the smaller of that quarter’s wages and the remaining annual allowance. The remaining allowance starts at the full annual taxable wage base, and falls as wages accumulate. Taxable wages are zero once the base is exhausted. Total and taxable wages are then summed across employees within each employer–year–quarter. The resulting dataset contains 11,015,554 employer–year–quarter observations with constructed quarterly total- and taxable-wage measures for each employer.

Merged data. The cleaned employer file is merged with the employee file aggregated at the employer–year–quarter level. A total of 8,802,019 employer–year–quarter observations are matched and retained for the analysis: they represent 69.27% of all distinct employer–year–quarter observations in the combined files. The following analyses use this matched sample.

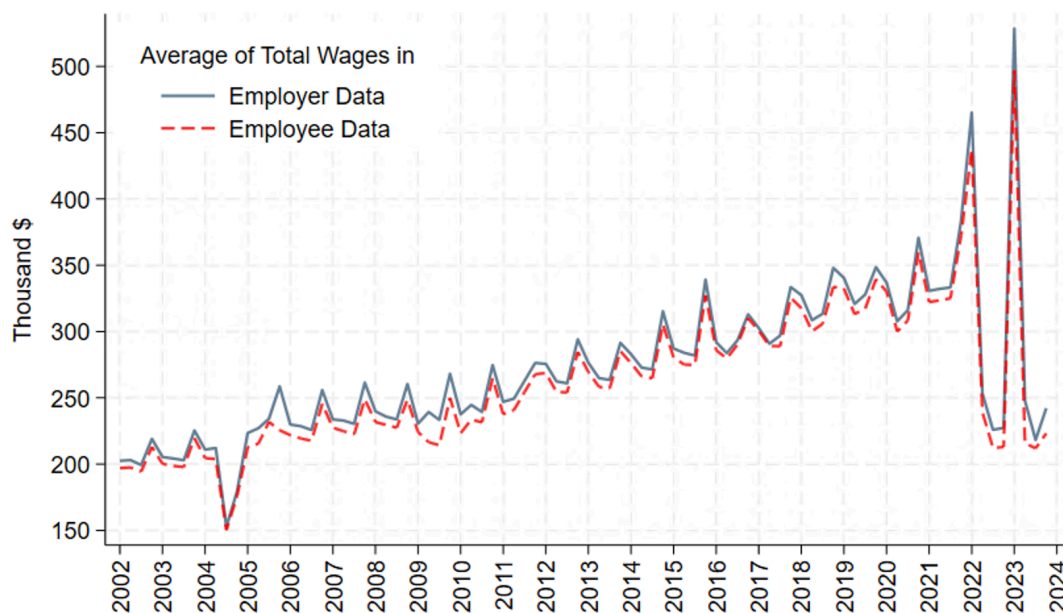
3 Issue: taxable wages equal total wages in the Employer data

Figure 1 compares average employer total wages (total_wage) and taxable wages (tax_wage) in the Employer data over time. The two series coincide from 2004 through 2015. Average taxable wages drop in 2016 and subsequently display clear peaks in the first quarter of each year, while average total wages remain on the same trend. This break suggests that tax_wage for 2004–2015 may not correctly measure taxable wages.



Two comparisons across the data sources provide additional evidence. Figure 2 compares average employer total wages provided in the Employer data (total_wage) with average employer total wages constructed from the wage files in the Employee data. The series are closely aligned, supporting the consistency of the total-wage measures across the two sources.

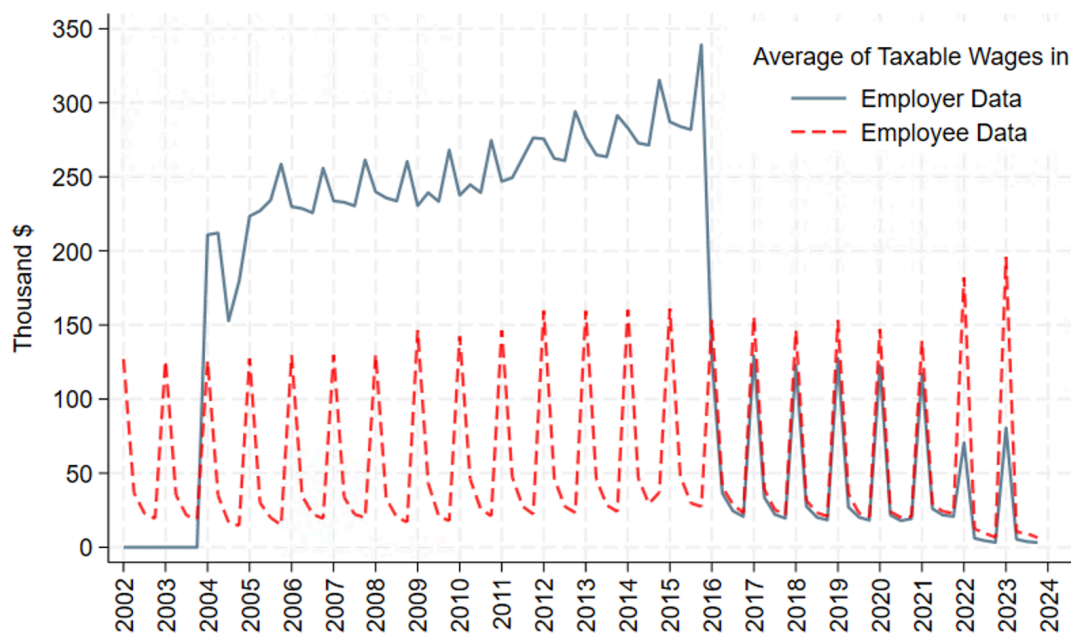
Figure 2. Average Employer Total Wages in the Employer and Employee Data



Note: The Employer-data series is the average of total_wage from the Employer data (Employer_Data_V1.txt) across employers for each year-quarter. The Employee-data series is the average, for each year-quarter, of employer total wages constructed from individual wage records in the Employee data (BU_UI_Employment.txt). The vertical axis is in thousands of dollars. The sample includes only employer-year-quarter observations matched across the two datasets.

Figure 3 compares average employer taxable wages in the Employer data (tax_wage) with taxable wages reconstructed from the wage files in the Employee data. The two measures of taxable wages differ between 2004 and 2015: the constructed measure varies over time and peaks in the first quarter, while the tax_wage variable from the Employer data, which always coincides with total wages during this period, is higher on average and substantially more stable. The two taxable-wage measures become more closely aligned from 2016 onward. The correlation between the two measures across matched employer-year-quarter observations from 2016 onward is 79% overall, 78% in Q1, 85% in Q2, 71% in Q3 and 95% in Q4. During the same period, the median absolute difference between the two variables is \$0.000061. The 75th percentile is \$0.40. The 95th percentile is \$7,000, or 16% of the average quarterly employer taxable wage of \$43,166. The correlation between the two measures increases to 81% when the sample is restricted to 2016–2021, consistent with the visual observation that discrepancies between the two variables are concentrated in Q1 of 2022 and 2023. Overall, this analysis reinforces the conclusion that tax_wage for 2004–2015 may be incorrect. Despite the discrepancies in later years, the agreement between the two series supports using reconstructed taxable wages from the Employee data as a substitute for tax_wage in 2004–2015.

Figure 3. Average Employer Taxable Wages in the Employer and Employee Data



Note: The Employer-data series is the average of `tax_wage` from the Employer data (`Employer_Data_V1.txt`) across employers for each year-quarter. The Employee-data series is the average for each year-quarter of employer taxable wages constructed from individual wage records in the Employee data (`BU_UI_Employment.txt`). The vertical axis is in thousands of dollars. The sample includes only employer-year-quarter observations matched across the two datasets.

4 Proposed solution

For analyses covering 2004–2015, do not use `tax_wage` from the Employer data. Instead, construct employee-employer-year-quarter taxable wages from the individual wage records in the Employee data and the relevant annual taxable wage base. Then sum workers' quarterly taxable wages by employer, year, and quarter.

ATTACHED REPLICATION CODE

The accompanying `do-file` constructs taxable wages from employee-level wage records using the applicable annual taxable wage base, creates the matched dataset, and reproduces the three figures. The `README` describes the inputs, output files, and run requirements. The log records the original run on 29 September 2026.